1 Introduction

Manual attendance in university classrooms wastes around ten minutes of every lecture and is easy to manipulate through proxy attendance, where one student marks another student as present.

This project, the Smart Attendance System, automates attendance using face recognition. A classroom camera captures frames, detects faces, matches them against enrolled students and records attendance in a central database that teachers can review from a web dashboard.

The main objectives are: to reduce attendance time to under one minute, to eliminate proxy attendance, to give teachers real-time reports, and to keep student biometric data secure.

2 Literature Review

Earlier systems used RFID cards, but cards can be shared between students so proxy attendance remains possible. Fingerprint scanners remove sharing but create queues because only one student can be scanned at a time.

Face recognition approaches based on Haar cascades were fast but inaccurate under poor lighting. Deep learning models such as FaceNet map each face to a 128-dimensional embedding where faces of the same person are close together, which gives far better accuracy in real classrooms.

We selected a FaceNet-style embedding approach with MTCNN for detection because published results show above 99 percent accuracy on the LFW benchmark while running in real time on a mid-range GPU.

3 System Architecture

The system follows a three-tier architecture. The capture tier runs on a classroom computer connected to a 1080p camera. The application tier is a Python Flask REST API that performs recognition. The data tier is a MySQL database that stores students, courses, sessions and attendance records.

The capture client sends one frame every two seconds to the API. The API detects faces with MTCNN, computes embeddings, and compares each embedding with enrolled embeddings using cosine similarity. A match is accepted when similarity is above a threshold of 0.75.

A student is marked present only after being recognised in at least three frames during the first fifteen minutes of the lecture, which prevents a photo held up briefly from counting as attendance.

4 Database Design

The database has five main tables: students, courses, enrollments, lecture_sessions and attendance. The attendance table references both the student and the lecture session and stores the first seen time and the number of frames in which the student was recognised.

Face embeddings are stored in a separate face_embeddings table encrypted with AES-256, and only the API service account can read them. Raw face images are deleted after enrollment so that only embeddings are kept, which reduces privacy risk.

Indexes on student_id and session_id keep attendance report queries under 100 milliseconds for a semester of data.

5 Face Recognition Module

Enrollment captures twenty images of each student from different angles. Each image is aligned, resized to 160 by 160 pixels and passed through the embedding model. The average embedding is stored as the student's template.

During recognition, each detected face is aligned the same way. Liveness detection checks eye blinking across consecutive frames to reject printed photos and phone screens.

We tuned the similarity threshold using a validation set of 40 students. A threshold of 0.75 gave a false acceptance rate of 0.5 percent and a false rejection rate of 3 percent.

6 Testing and Results

Unit tests cover the API endpoints and the matching logic. Integration tests replay recorded classroom videos through the full pipeline.

In a pilot with three classes of 45 students over two weeks, the system marked attendance in an average of 40 seconds per lecture with 96.8 percent accuracy compared with manual roll calls.

Most errors came from students wearing masks or sitting at the back row in poor lighting. Adding a second camera at the back of the room reduced these errors by half.

7 Conclusion and Future Work

The Smart Attendance System reduced attendance time from about ten minutes to under one minute and removed proxy attendance in the pilot classes.

Future work includes masked face recognition, a mobile app for teachers, and integration with the university learning management system.